

1. Write a Pandas program to detect missing values of a given DataFrame. Display True or False.

Code:

```
import pandas as pd
import numpy as np

data = {
    'A': [1, 2, np.nan, 4],
    'B': [5, np.nan, 7, 8],
    'C': [9, 10, 11, np.nan]
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

print("\nMissing Values (True/False):")
print(df.isnull())
```

Output:

```
Original DataFrame:
   A    B    C
0  1.0  5.0  9.0
1  2.0  NaN 10.0
2  NaN  7.0 11.0
3  4.0  8.0  NaN

Missing Values (True/False):
   A    B    C
0  False False False
1  False  True False
2   True False False
3  False False  True
```

2. Write a Pandas program to find and replace the missing values in a given DataFrame which do not have any valuable information.

Code:

```
import
pandas as pd
import
numpy as np

data = {
    'A': [1, np.nan, 3, 4],
    'B': [5, 6, np.nan, 8],
    'C': [np.nan, 10, 11, 12]
}

df = pd.DataFrame(data)

print("Original
DataFrame:") print(df)

df.fillna(0, inplace=True)

print("\nDataFrame after replacing missing values:")
print(df)
```

Output:

```
Original DataFrame:
   A    B    C
0  1.0  5.0  NaN
1  NaN  6.0  10.0
2  3.0  NaN  11.0
3  4.0  8.0  12.0

DataFrame after replacing missing values:
   A    B    C
0  1.0  5.0  0.0
1  0.0  6.0  10.0
2  3.0  0.0  11.0
3  4.0  8.0  12.0
```
